

PEP PERFORMANCE REPORT

Flow Analysis | Data Pipeline Methodology

Date:	14 July 2026	Project:	Global Events, Commodities and Currency Markets — DSB Group Project
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1. Summary

This is the final, closeout Performance Report for the DSB Group Project, superseding the two interim reports dated 2026-05-20 (Part I complete) and 2026-06-08 (Part II complete, Sprint 3 in progress). The project leveraged a Data Pipeline methodology to analyse five datasets: Brent Oil (1946–2026), Gold (1915–2026), Silver (1915–2026), the SWIFT Global Currency Tracker (January–April 2026), and USDG/Global Dollar stablecoin (April 2025–April 2026). The analysis covers the 2026 US–Iran conflict, the Hormuz shipping crisis, energy supply shocks, central bank gold accumulation, and BRICS bloc involvement in non-dollar settlement systems.

As of this report date, all core assignment work is complete and the final submission was made: Part I (30 questions + 6 bonus), Part II (16 questions), the 8–10 page Final Report (with extended appendix). In addition, the team delivered an extended, unplanned forecasting pipeline - four tiers of econometric and machine-learning models (ARIMA/GARCH, Random Forest/LightGBM/LSTM/BiGRU, VECM/Markov-regime-switching/USDG adoption modelling, and a Streamlit interactive dashboard) plus an out-of-sample validation against May–June 2026 actuals. This extended work was beyond the original scope and hour budget and is treated separately in the variance analysis below (Section 8). The project is now closed. This report documents final schedule performance, final resource utilisation, consolidated key findings, resolved risks, and retrospective recommendations for future team projects.

2. Project Scope and Objectives

Scope

Analyse and compare commodity price trends and global currency indicators using a Data Pipeline methodology, connecting observed data patterns to major global events including the Hormuz crisis, energy supply disruptions, gold accumulation, sanctions, BRICS settlement discussions, and currency market shifts.

Datasets in scope:

- Brent Oil price history - 963 monthly observations, 1946–2026
- Gold price history - 1,336 monthly observations, 1915–2026
- Silver price history - 1,336 monthly observations, 1915–2026
- SWIFT Global Currency Tracker - 330 rows, 8 metrics, Jan–Apr 2026
- USDG (Global Dollar stablecoin) - 366 daily observations, Apr 2025–Apr 2026 (collected via CoinGecko API)

Extended scope (delivered beyond original plan): FRED macro series (VIX, DXY, TIPS) for the forecasting pipeline; out-of-sample actuals (yfinance, May–June 2026) for forecast validation.

Objectives

- Collect and integrate all relevant datasets into a standardised, reproducible pipeline.
- Cleanse, transform, and validate data to ensure analytical integrity.
- Generate descriptive, exploratory, predictive, and prescriptive analytics across four analytical layers.
- Produce comparative visualisations linking commodity and currency movements to global events.
- Deliver actionable, evidence-based insights written for non-technical stakeholders.
- Validate descriptive/market-research findings against formal statistical and machine-learning forecasting models.

3. Performance Analysis

A. Work Completed This Period (since the 2026-06-08 report)

- Completed Final Report writing across all sections (Introduction, Data Overview, Part I Findings, Part II Findings, Final Recommendation, Appendix), incorporating all 30+16+6 answered questions and chart references, finished 2026-07-01.
- Built the Excel Spreadsheet Workbook (Raw Data, Cleaned Data, Annual Averages, SWIFT Analysis, Correlation Table sheets).
- Conducted full QA review of the report: grammar, chart labelling, axis titles, and APA-style citations.
- Finalised the shared Google Drive folder, Trello board, and GitHub repository for the project archive.
- Submitted the complete deliverable package on 2026-07-12, report, Jupiter Notebook, Power BI, presentation, and supporting charts.
- Delivered an extended, four-tier forecasting pipeline as bonus/portfolio work: Tier 1, stationarity testing (ADF/KPSS), structural break detection (CUSUM + PELT), ARIMA/ARIMAX walk-forward forecasting, GARCH/EGARCH volatility modelling, Diebold-Mariano significance testing; Tier 2, Random Forest (500 trees), LightGBM, LSTM(64) across three assets, BiGRU(32) for Gold; Tier 3, Gold–Silver Johansen cointegration test + VECM, Markov regime-switching model for Brent, USDG peg-deviation and adoption modelling; Tier 4, interactive Streamlit dashboard with animated price charts, model comparison view, and macro-context overlay (VIX/DXY/TIPS).
- Validated forecasts against real May–June 2026 market actuals (yfinance). 6 of 7 data points landed within the 95% confidence interval.
- Produced a full academic-style forecasting report and a 16-slide Marp presentation deck documenting the extended work.
- Resolved several technical issues discovered during the extended build (see Section 7).

B. Work Planned for Next Period

- None — the core deliverable package (report, Excel workbook, presentation) is submitted, all collaboration tooling (Google Drive, Power BI, Trello, GitHub) is finalised, and the project is closed.

C. Key Accomplishments

- Full analytical coverage across the entire project lifecycle: All 30 + 16 + 6 core/bonus questions answered — no questions skipped or deferred, plus the entire extended forecasting pipeline, none of which was contractually required.
- On-time, complete submission: Final deliverable package (report + Excel workbook) submitted 2026-07-12, closing out the project.
- Independent data acquisition: USDG dataset (366 days) collected and integrated via CoinGecko public API with full reproducibility documentation.
- Key market finding, Hormuz crisis: Brent Oil moved from a \$102.86/barrel January 2026 baseline to a ~\$126/barrel peak, a 22% supply-shock spike, the largest short-term movement in the 80-year dataset.
- Key market finding, Gold: All-time intraday high of \$5,595/oz reached 29 January 2026 (one month before the Hormuz military escalation), with a 2026 annual average of \$4,881.62, confirming the move was structural (central bank accumulation), not purely crisis-reactive.
- Key currency finding: USD retains global payment dominance at 51.14% (April 2026) but its reserve share has declined from 72% to 56.92% over 20 years; CNY rose to SWIFT Rank 5 and BRICS Pay reached \$4B/month operational volume.

- Extended forecasting results: Gold's four best-performing models (LSTM, ARIMA, BiGRU, VECM) clustered tightly around 3.9% MAPE , LSTM marginally ahead at 3.88% vs. ARIMA's 3.90%, a 0.02pp gap read as a practical tie rather than a decisive winner; Brent's best was Random Forest(500) at 6.49% MAPE, only marginally ahead of the naive random-walk baseline (6.50%); no model meaningfully beat the naive baseline for Silver; Gold–Silver cointegration was formally rejected (Johansen rank = 0) despite the 0.95 correlation; Brent's stress-regime variance was ~680× its calm-regime variance; USDG's peg deviation was just 6.8×10^{-5} USD (near-perfect peg stability).
- Out-of-sample validation: 6 of 7 forecasted May–June 2026 values fell within the models' 95% confidence intervals, giving external, post-hoc confidence in the pipeline's calibration.

4. Schedule Performance

Milestone	Planned Date	Actual Date	Status	Notes
Project Kickoff	Week of 2026-04-28	2026-04-28	✓ Complete	Assignment analysis + data review initiated
Data Pipeline Design	Week of 2026-05-05	2026-05-07	✓ Complete	Architecture defined; Collect -> Clean -> Transform -> Analyze -> Visualize -> Report
Data Extraction & Cleaning	Week of 2026-05-12	2026-05-15	✓ Complete	All 5 datasets loaded, cleaned, and validated
Preliminary Analysis (Part I)	Week of 2026-05-19	2026-05-19	✓ Complete	All 30 + 6 bonus questions answered
In-depth Analysis & Reporting (Part II)	Week of 2026-05-22	2026-05-22	✓ Complete	All 16 market research questions answered
Final Report Writing & QA	Internal target 2026-06-22	2026-07-01	✓ Complete	Report, chart insertion, and citation QA finished ahead of submission
Extended Forecasting Pipeline (Tier 1–4, unplanned bonus scope)	Not in original plan	2026-06-21	✓ Complete	Delivered as portfolio-quality bonus work; see Section 8 for budget impact
Google Drive / Trello / GitHub Archival	Ongoing	2026-07-10	✓ Complete	All collaboration and archival tooling finalised before submission

Schedule note: All phases were completed, and the project is now closed.

5. Budget and Resource Utilization

Resource	Planned (hrs)	Actual (hrs)	Variance	Status	%
Data Engineering & Analysis (Abdulhadi Kirkici + Sagar Vig)	30	42	+12	✓ Complete	140%
Data Researchers (Parvesh Kumar + Sameep Santosh Dokhale)	20	19	-1	✓ Complete	95%
Report Writer (Tabassum)	15	17	+2	✓ Complete	113%
Software Tools	0	0	0	✓ All free	—
Total	65	78	+13	✓ Project closed	120%

Variance driver: The +12 hr overage on Data Engineering & Analysis covers both final report/Excel workbook completion and the extended forecasting pipeline (Tier 1–4 + Streamlit dashboard) - the latter ran a few hours past its own informal estimate, which accounts for most of this line's variance. Researchers and Report Writer landed close to plan once report writing and final review were folded in.

Tools used (all free):

- Python 3 + Jupyter Notebook: data pipeline and all analytics
- pandas, numpy, matplotlib, seaborn, scikit-learn: analysis and visualisation libraries
- statsmodels, arch, lightgbm, tensorflow/keras: extended forecasting pipeline (Tier 1–3)
- Streamlit : Tier 4 interactive dashboard
- Power BI : dashboard
- Google Docs : report drafting
- Trello : task management
- WhatsApp: team communication

6. Quality and Technical Performance

Quality Indicator	Details
Data Accuracy	All five core datasets verified: 0 missing values in Brent/Gold/Silver; SWIFT 388 NaN confirmed as structural; 1 USDG duplicate date resolved via <code>drop_duplicates(keep="first")</code> . Silver dataset's March/April 2026 rows were identified as carrying an identical stale value (\$88.36); this was documented rather than corrected, as the source file could not be re-pulled.
Pipeline Uptime	100% across both the core notebook and the extended Tier 1–4 forecasting notebooks — all executed end-to-end without unresolved runtime errors. A <code>to_timestamp('MS')</code> incompatibility with the installed pandas version was identified and fixed ('MS' period-frequency argument no longer accepted).
Extended Pipeline Fixes	Streamlit deprecation (use <code>container_width=True</code> → <code>width='stretch'</code>) resolved for Streamlit ≥1.37. Arrow/PyArrow serialisation issue in the metrics table resolved by replacing '-' placeholder strings with <code>None + pd.to_numeric(errors="coerce")</code> .
Compliance	All data handling in line with data source requirements. CoinGecko Public API free tier terms observed: data used for academic research, attribution included, no commercial redistribution. Full data collection methodology and reproducible Python code documented for academic integrity. Citations included for all external data sources and academic references used in both the core report and the extended forecasting appendix.

7. Risks and Issues

Risk / Issue	Status	Resolution
Data Gaps — USDG launch period (Nov 2024–Apr 2025) unavailable	✅ Resolved / Accepted	CoinGecko free tier limits historical access to 365 days; documented as an explicit limitation. 366 days proved sufficient for all analytical objectives.
Pipeline Integration Bug — Date parsing (MM/DD/YYYY vs DD/MM/YYYY)	✅ Resolved	Explicit format string applied to all commodity date columns; verified via min/max sanity checks. No data loss.
Pipeline Integration Bug — <code>to_timestamp('MS')</code> incompatibility	✅ Resolved	Replaced with <code>to_timestamp()</code> ; newer pandas versions no longer accept the 'MS' period-frequency argument.
Silver data quality — stale duplicate value (Mar/Apr 2026)	✅ Documented / Accepted	CSV shows an identical \$88.36 value for both months; flagged in the report rather than silently corrected, since the true April 2026 value (~\$75.96, per external reference) was not part of the original assignment dataset.
Team availability variance — technical workload concentrated on one person	✅ Closed / Noted for retrospective	The extended forecasting pipeline (Tier 1–4) was carried almost entirely by Abdulhadi Kirkici as self-directed bonus work, on top of his core Data Engineering & Analysis hours. This was a deliberate choice, not an imposed imbalance, but is flagged in Section 10 for future project planning.
Final report coordination — Google Drive, Trello, GitHub archival	✅ Resolved	Shared Google Drive folder, Trello board, and GitHub repository all finalised (2026-07-10) ahead of submission; all deliverables archived.

8. Variance Analysis

Area	Analysis
Schedule	All core-assignment milestones (data pipeline, cleaning, Part I, Part II, final report) were completed on or ahead of the team's internal sprint schedule.
Budget (Hours)	Final hours came in at 78 of 65 planned hours (120% utilisation) — a modest, expected overage rather than a major blowout. Most of it sits in Data Engineering & Analysis, where the extended Tier 1–4 forecasting pipeline ran a few hours past its own informal estimate on top of the core report/Excel/presentation work. This is a positive variance: it reflects additional value delivered (portfolio-quality forecasting pipeline, validated against real market outcomes, submitted ahead of deadline), not inefficiency or rework on the required deliverable.

9. Forecast

The project is closed. Every planned and bonus deliverable has been completed and submitted, three days ahead of the instructor-confirmed deadline. Only one item remains, and it is outside the team's control:
Final deliverable status:

Deliverable	Status	Owner	Date
Final Report (8–10 pp + extended appendix)	✔ Submitted	All team	2026-07-12
Jupiter Notebook	✔ Submitted	Abdulahdi Kirkici	2026-07-12
Extended Forecasting Report + Slides (bonus)	✔ Complete	Abdulahdi Kirkici	2026-06-21
Google Drive / Trello / GitHub Archival	✔ Complete	Tabassum Nadavinmani	2026-07-10

10. Recommendations

1. Scope "bonus" work explicitly, even if the overrun ends up small. The extended forecasting pipeline ran a few hours past its own informal estimate and was carried by one team member on top of the core assignment. The overage was modest, but agreeing upfront whether self-directed extensions count against shared team hours or an individual's own initiative avoids ambiguity in the budget table.

Authorizations

Team Leader	Co-Team Leader
Signature: _____ Name: Tabassum Nadavinmani Date: 14 July 2026	Signature: _____ Name: Abdulhadi Kirkici Date: 14 July 2026